INCH-POUND
A-A-59484
14 March 2001
SUPERSEDING
MIL-B-23755D
29 March 1991

COMMERCIAL ITEM DESCRIPTION

BRAKE MACHINES, SHEET METAL, MANUALLY OPERATED STANDARD AND BOX-PAN TYPE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers two types (standard and box-pan) of manually operated brake machines which are used for bending mild steel sheet metal ranging from 22 through 12 USS gauge. Machines covered under this CID will have the capacity to handle work piece lengths ranging from 2 through 12 feet. In addition, the scope includes machines capable of forming box-pans to a depth ranging from three through 8 inches.

2. CLASSIFICATION. The brake machines shall be of the following types, classes, and sizes listed in tables I and II. The type, class, and size to be furnished shall be as specified (see 7.3(b)).

Type I - Standard brakes.

Class 1 - Floor mounted.
Class 2 - Bench mounted.

Size - (see table I for class 1 and 2 sizes and capacities)

Type II - Box-pan brakes.

Class 1 - Floor mounted.
Class 2 - Bench mounted.

Size - (see table II for class 1 and 2 sizes and capacities)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3441

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:16Z
Check the source to verify that this is the current version before use.

TABLE I. Type I, standard brakes, sizes, and capacities.

Class 1 - Floor mounted			Class 2 - Bench mounted		
	Capacity			Capacity	
Size	Work piece bending length (ft)	Work piece thickness (USS gauge)	Size	Work piece bending length (ft)	Work piece thickness (USS gauge)
01	4	12	01	2	16
02	4	14	02	3	16
03	4	16	03	4	16
04	4	22	04	4	22
05	6	12			
06	6	14			
07	8	12			
08	8	14			
09	8	16			
10	8	18			
11	10	12			
12	10	14			
13	10	16			
14	10	18			
15	12	18			

NOTE: Machine sizes and capacities shall not be less than the stated requirements of size ordered, nor greater than the capacities of the next larger size shown in table I for classes 1 and 2.

TABLE II. Type II, box-pan brakes, sizes, and capacities.

Class 1 - Floor mounted				Class 2 - Bench mounted			
	Capacity				Capacity		
Size	Work piece bending length (ft)	Work piece thickness (USS gauge)	Box-pan depth (in)	Size	Work piece bending length (ft)	Work piece thickness (USS gauge)	Box-pan depth (in)
01	2	18	3	01	2	16	3
02	4	12	4	02	3	16	3
03	4	12	6	03	4	16	4
04	4	12	8	04	3	22	3
05	4	14	6				
06	4	16	4				
07	4	22	3				
08	6	12	4				
09	6	12	6				
10	6	12	8				
11	6	16	4				
12	8	12	6				
13	8	12	8				
14	8	14	6				
15	8	14	8				
16	8	16	4				
17	10	12	6				
18	10	12	8				
19	10	14	6				
20	10	14	8				
21	10	16	6				

NOTE: Machine sizes and capacities shall not be less than the stated requirements of size ordered, nor greater than the capacities of the next larger size shown in table II for classes 1 and 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:16Z
Check the source to verify that this is the current version before use.

3. SALIENT CHARACTERISTICS

3.1 Materials and composition. The machine shall be new (not a prototype) and one of the manufacturer's current models capable of operations in accordance with the performance requirements in this description. All machines shall include all components, parts, and features necessary to meet the performance specified. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair. The machine shall be manually operated and designed to accomplish bending. Type I machines will be capable of forming straight bends. Type II machines will be capable of performing all type I performance requirements and be equipped with removable ancillary parts capable of forming multiple intersecting bends such as those required when forming boxes, pans, and cabinets.

3.2 Performance of type I - standard brakes. The machine shall be capable of bending mild steel of not less than 55,000 pounds per square inch (PSI) tensile strength in work piece sizes up to the rated capacity for the machine specified bends. These operations shall be performed to the accuracy and requirements stated in 3.2.1 through 3.2.11.

3.2.1 Angle of bend. All machines shall be capable of forming bends within the range of 0 degrees to 135 degrees, measured from the horizontal plane (180 degrees to 45 degrees including angle).

3.2.2 Bending components. All bending components shall have a pivot range that meets or exceeds the angle of bend requirement of paragraph 3.2.1. Reinforcement components, such as truss rods, shall have a means of tension adjustment.

3.2.3 Bending accuracy. The machine shall be capable of forming 90 degree bends to an accuracy of ± 2 degrees.

3.2.4 Repeatability. The machine shall be capable of making repetitive bends to an accuracy of ± 2 degrees by using stopping devices.

3.2.5 Bend radius. The machines shall be capable of making sharp bends having an inside radius not more than twice the thickness of the metal being bent. The machine shall also be capable of bending larger radii by making "freehand" multiple partial bends, or by using specifically radiused nose bars.

3.2.6 Narrow flanges. The machine shall be capable of forming full-length flanges having a width of 1 inch or wider. The flange shall be uniform in height within ± 0.15 inch for the full length of the work piece.

3.2.7 Base. The base shall have adequate strength and rigidity to maintain all parts in alignment when bending sheet metal of full capacity (length, thickness, and depth) as specified in tables I and II. A means shall be provided for securely fastening the base to the floor or foundation for class1 brakes, and to a bench or stand for class 2 brakes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:16Z
Check the source to verify that this is the current version before use.

3.2.8 Handles. The machine shall have all required handles to allow the operator to apply pressure evenly throughout a full capacity sheet metal work piece. Handles and other movable parts shall have adequate strength and rigidity to withstand the applied forces required to bend sheet metal of full capacity (length, thickness, and depth) as specified in tables I and II.

3.2.9 Reverse bends. All brakes shall have the capability for minimum reverse bends of 1/4 inch.

3.2.10 Clamping devices. Adjustable and adequate clamping devices shall be provided to securely hold, the work piece in place without slippage during bending as specified in tables I and II.

3.2.11 Counterweights. All counterweights shall be removable and have a means of adjustment on their supports.

3.3 Performance of type II - box-pan brakes. Type II box-pan brakes shall be capable of all performance requirements for type I standard brakes. Type II brakes shall accommodate removable and rearrangeable fingers (or other means to perform multiple bends) extending across the entire work piece length of bend. The fingers shall be fully supported and securely held in position during bending operations. Additional performance requirements for type II brakes are stated in paragraphs 3.3.1 through 3.3.3.

3.3.1 Fingers. A complete set of removable, rearrangeable, and random width fingers for forming sharp bends across the entire work piece length shall be furnished with each brake. A means to permanently identify the width of each finger in inches and fractions of an inch shall be provided. All adapters, screws, bolts, and nuts for mounting and aligning the fingers shall be furnished with each brake. The finger sets shall form bends of lengths and increments as follows:

a. Type II, class 1: All sizes - 3 inches to the full bending length in 1-inch increments as specified in table II.

b. Type II, class 2: All sizes - 3/4 inch to the full bending length in 1/4-inch increments as specified in table II.

3.3.2 Multiple bends. All type II box-pan machines shall be capable of forming multiple bends having previously formed side flanges.

3.3.3 Tubular shapes. All type II box-pan machines shall be capable of forming tubular shapes.

3.4 Controls. All mechanical operating controls shall be located convenient to the operator's work station.

3.5 Assembly, operating, and maintenance requirements. All brake machines procured under this CID shall include adequate manuals and/or instructions to accomplish assembly, operations, and maintenance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:16Z
Check the source to verify that this is the current version before use.

3.6 Optional equipment, guidance, tools and supplies. All optional equipment, guidance, tools, and supplies which are available, but not customarily furnished with the machine, shall be described and priced with all responses to solicitations under this CID. Available items that are furnished and priced with the machine need not be described and priced separately. Categories of items to be included, but not limited to, are as follows:

a. Equipment: Attachments and auxiliary equipment such as benches for class 1 and machines, worklights, radius former bars, tinner's mould, fingers, etc. shall be described and priced with all responses to solicitations under this CID.

b. Guidance: Guidance material for assembly, operations, safety, and maintenance.

c. Tools and supplies: Special tools, such as wrenches, and supplies, such as lubricants.

4. REGULATORY REQUIREMENTS

4.1 Health, safety, and environment. The machine shall adhere to all Federal, state, and local health, safety, and environmental regulations. Covers, guards, or other safety devices normally furnished as standard on the manufacturers commercially supplied machines shall be provided. Safety devices shall not interfere with machine operation and shall allow access to guarded parts. No environmentally prohibited material or components shall be used in the manufacture, finishing, or packing of the machines.

4.2 Recycled materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

5.2.1 The supplier must be able to show test data or lab results of meeting the salient characteristics.

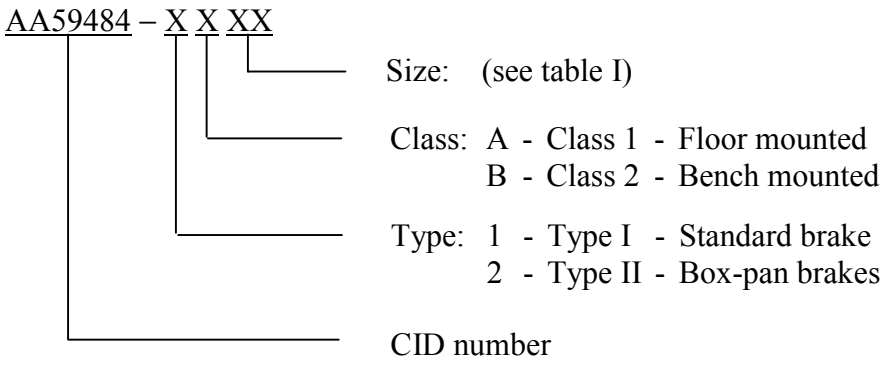
5.2.2 The supplier must provide a warranty of replacing defective items.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 7.3(f)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:16Z
Check the source to verify that this is the current version before use.

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



EXAMPLE: AA59484-2A09 would be for a brake machine as follows:

A floor mounted box-pan machine for sheet metal up to 6-feet long and up to 12-gauge thickness. The machine would also have the capability to form a 6-inch deep box.

7.2 Source of documents.

7.2.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.3 Ordering data. Acquisition documents must specify the following:

a. CID document number, revision, and CID PIN.
b. Product classification (type, class, and size) (see 2 and tables I and II).
c. Quantity of machines required.
d. Any special uses or applications.
e. Any special requirements.
f. Packaging requirements (see 6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:16Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

Custodians:
Army - AR
Navy - SH
Air Force - 99

Reviewer:
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FEE

Preparing activity:
DLA - GS5

(Project 3441-0192)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:16Z
Check the source to verify that this is the current version before use.